SOFTWARE ARCHITECTURE

Overview of software architecture

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Enterprise Architecture Concepts

- Why do we need Enterprise Architecture?
- Knowing the role of an architect
- Commonly known EA Frameworks
- Architecture segregation
- Summary

Why do we need Enterprise Architecture?

- It was confusing to understand the role of an enterprise architect
- Many architectural roles and terms, such as architect, solution architect, enterprise architect, data architect, blueprint, system diagram, and so on
- In general, an IT architect role is to draw a few boxes with a few suggestions; the rest is with the development community.
- The architect role is quite easy, just drawing a diagram and not doing anything else.
- Agree or disagree? And Why?

Why do we need Enterprise Architecture?

- Definition of Enterprise Architecture
  - In simple terms, an enterprise is nothing but human endeavor.
  - Where people are collaborating for a particular purpose supported by a platform.

Why do we need Enterprise Architecture?

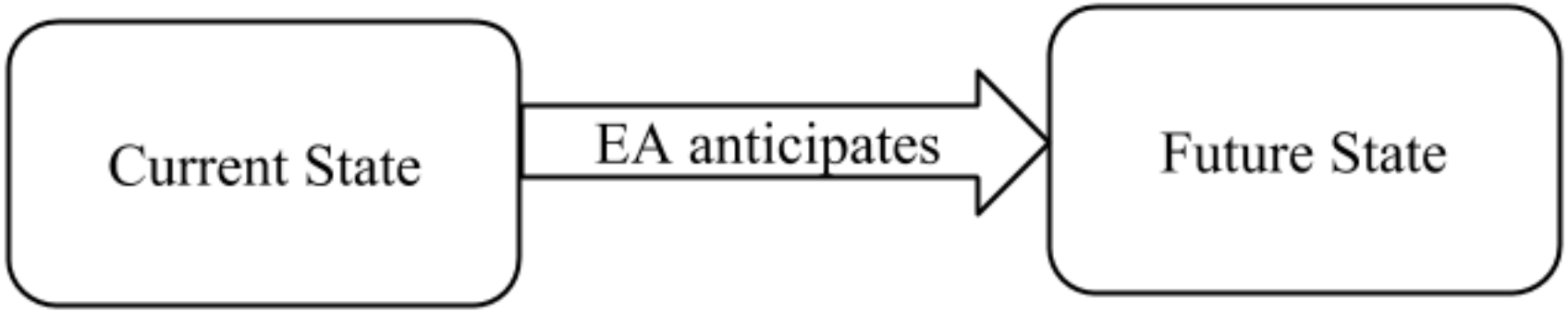
- Definition of Enterprise Architecture
  - An example of an online e-commerce company.
    - Employees work together to produce the firm's profits using their various platforms, such as infrastructure, software, equipment, building, and so on.
    - Enterprise has the structure/arrangements of all these pieces/components to build the complete organization.
    - This is the exact place where enterprise architecture plays its key role. Every enterprise has an enterprise architect.

Why do we need Enterprise Architecture?

- Definition of Enterprise Architecture
  - EA is a process of architecting that applies the discipline to produce the prescribed output components.
  - This process needs experience, skill, discipline, and descriptions. Consider the following image, where EA anticipates the system in two key states:

Why do we need Enterprise Architecture?

- Definition of Enterprise Architecture
  - EA anticipates the system in two key states:



Why do we need Enterprise Architecture?

- Every enterprise needs an enterprise architect
- An example
  - You need a car for business activities, you have two choices, either drive yourself or rent a driver.
  - Still, you will need the driving capability to operate the car. EA is pretty like this.
- As depicted in the preceding diagram, EA anticipates the system in two key states, which are as follows:
  - How it currently is
  - How it will be in the future

Why do we need Enterprise Architecture?

- They work on options/alternatives to move from the current to a future state of an enterprise system.
- Enterprise Architecture does the following:
  - Creates the frameworks to manage the architecture
  - Details the descriptions of the architecture
  - Road maps to lay the best way to change/improve the architecture
  - Defines constraints/opportunities
  - Anticipates the costs and benefits
  - Evaluates the risks and values

Stakeholders of Enterprise Architecture

- Enterprise Architecture is so special because of its holistic view of management and the evolution of an enterprise holistically.
- It has a unique combination of specialist technologies, such as architecture frameworks and design pattern practices.

Stakeholders of Enterprise Architecture

- A special EA has the following key stakeholders/users in its ecosystem:

S.No.	Stakeholders	Organizational actions
1	Strategic planner	Capability planning Set strategic direction Impact analysis
2	Decision makers	Investment Divestment Approvals for the project Alignment with strategic direction
3	Analyst	Quality assurance Compliance, Alignment with business goals
4	Project managers	Solution development Investigate opportunities Analysis of existing options

Business benefits

- Many organizations intervened without EAs
- It is better to architect before creating any system.
- In terms of business benefits, cost is the key factor in the meaning of Return on Investment (RoI).
- EA has the opportunity to prove its value for its own stakeholders with three major benefits, ranging from tactical to strategic positions. They are as follows:
  - Cost reduction by technology standardization
  - Business Process Improvement (BPI)
  - Strategic differentiation

Business benefits

- Technology standardization adds a lot of efficiency to create indirect benefits.
- The variety of technologies and products were built to serve the business purpose due to historical acquisitions and mergers.
- All businesses have processes
  - There are people involved with few steps for the system to get things done.
  - During rapid business growth, the processes become chaotic, which leads to duplicate efforts across departments.
  - In turn, stakeholders do not leverage the collaboration and cross learning.

Business Process Improvement (BPI)

- An industry approach that is designed to support the enterprise for the realignment of the existing business operational process into the significantly improved process.
- It helps the enterprise to identify and adopt in a better way using industry tools and techniques.
- Strategic Differentiation efforts make a firm create the perception in customers minds of receiving something of greater value than is offered by the competition.
- An effective differentiation strategy is the best tool to highlight a business's unique features and make it stand out from the crowd.

Knowing the role of an architect

- In the IT industry, two higher-level architects are named as follows:
  - Solution architect (SA)
  - Enterprise architect (EA)

Knowing the role of an architect

Enterprise Architecture is a much broader discipline than Solution Architecture

Enterprise Architecture combines of

Business Architecture	Application Architecture,	Data Architecture	Technology Architecture

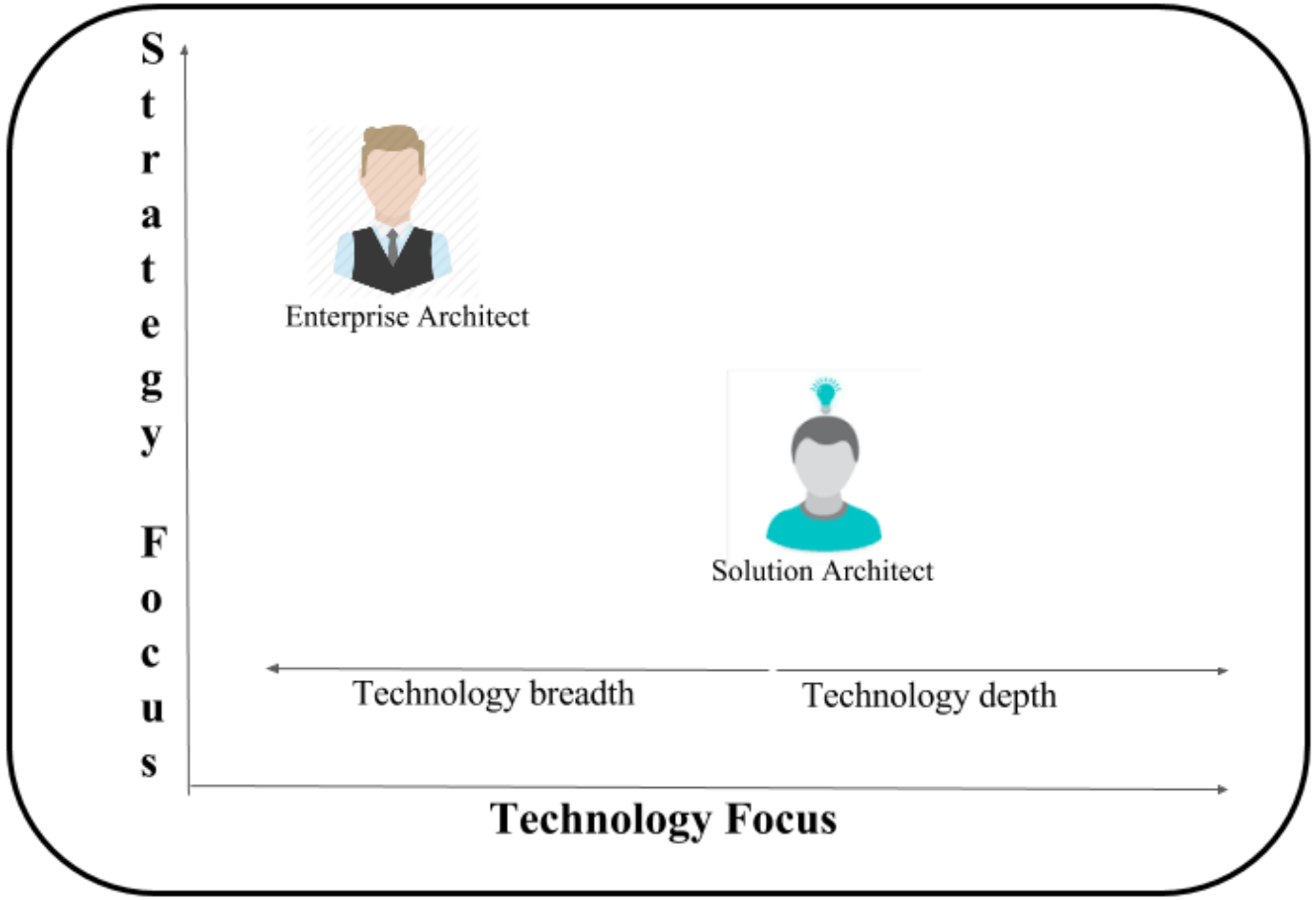
Solution architect (SA)

- SA is focused on a specific solution and addresses the technological details that are compiled to the standards, roadmaps, and strategic objectives of the business.
- In comparison with SA, EA is a more senior level.
- EA takes a strategic, inclusive, and long-term view of goals, opportunities, and challenges facing the company.
- However, SA is assigned to a particular project/program in an enterprise to ensure technical integrity and consistency of the solution at every stage of its life cycle.

Role comparison between EA and SA

- SA role for an Internet based telephony system
  - Build tools, such as code generation, automation, and so on around the existing telephony system.
  - Need the skill set to understand the existing system, and then provide better solutions to improve the productivity and performance of the existing ecosystem.
  - Like an individual contributor to building effective and efficient solutions to improvise the current system.
- EA role for a leading financial company.
  - Build the enterprise data hub using emerging big data technology.

Role comparison between EA and SA



Scope of EA and SA

- The SA scope is limited within a project team and the expected delivery is to make the system quality of the solution for the business.
- The EA scope is beyond SA by identifying or envisioning the future state of an organization.
- EA is superior to SA. EA has the vision of end-to-end broader system knowledge; but SA is bound to a specific problem statement.
- In terms of enterprise role, EA role is close to Chief Architect, whereas SA is at the Senior Architect level.

Commonly known EA Frameworks

- The Enterprise Architecture Framework (EAF) inspires software development processes in the industry.
- It is essential to fulfill the mission of the associated enterprise.
- EA serves as the blueprint for the system and the project that develops it.
- An EAF can describe the underlying infrastructure, thus providing the groundwork for the hardware, software, and networks to work together.

Discussion

- What is “blueprint” mean?
- Could you find other words that have a similar meaning to “blueprint”?

Discussion

- A model
- A plan
- A prototype
- A design
- A scheme
- A sketch
- …

EA Frameworks

- With EAF, the organization will be in a situation to understand and analyze the weaknesses or inconsistencies to be identified and addressed.
- As per the fundamentals of computing, a framework is often a layered structure indicating what kind of programs can or should be built, and how they will interrelate.
- A few established EAFs available in the industry today. Some of them were developed for a very specific area, whereas others have a broader coverage with complete functionality.

EA Frameworks

- There are two common types of EAFs used in the industry, which are as follows:
  - General Purpose Framework
  - Domain Specific Framework

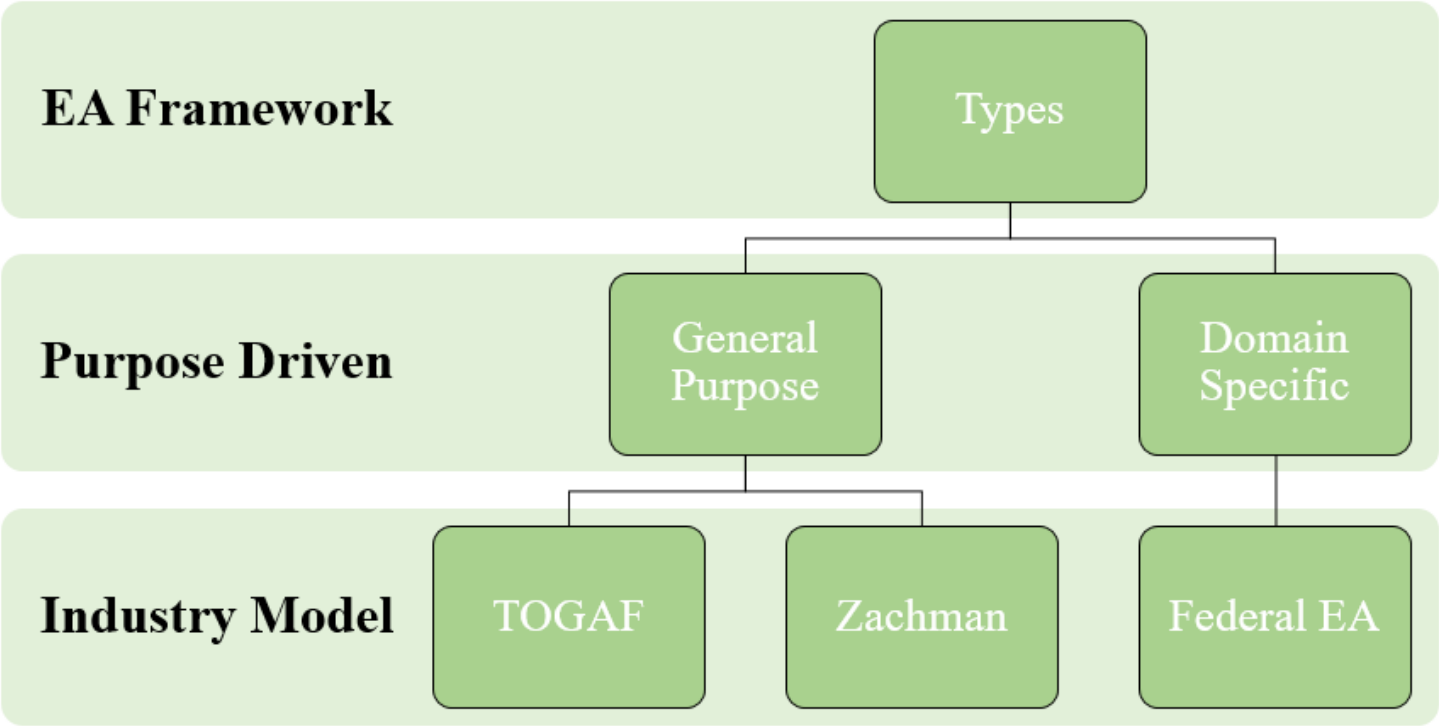
EA Frameworks

- General Purpose Framework
  - These frameworks are designed by being agnostic to any specific implementation.
  - They have no specific business drivers in terms of an enterprise specific scenario, but rather, they are capability based.
  - Some of the well- known general purpose EA Frameworks are TOGAF and Zachman.
- Domain Specific Framework

EA Frameworks

- Domain Specific Framework
  - These frameworks are derived from the common EA effort, in turn referred to as domain specific.
  - They are derived with a predefined set of business conditions and concerns because they may have originated from an Enterprise Architecture team or process improvement effort.
  - On rolling out to the industry, these frameworks are mostly driven by government agencies or other geographies.

EA Frameworks



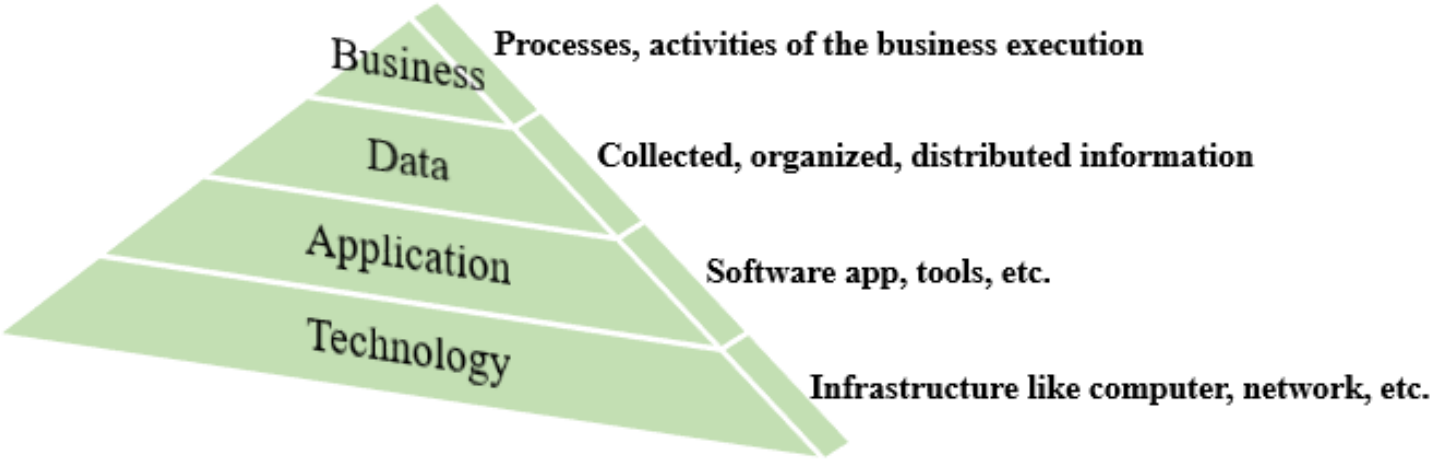
EA Frameworks

- By design, the EA Framework provides a conceptual framework to explain the following:
  - How the key terms are related to each other conceptually for architectural description
  - The following are the number of scenarios for the enterprise architectural activities during the software life cycle:
    - Evolutionary system
    - Existing architecture
    - Architectural evaluation
  - The role of the stakeholders in the creation and use of an architecture description

Architecture segregation

- In 1992, Steven H. Spewak defined Enterprise Architecture Planning (EAP) as the process of defining architectures for the use of information in support of the business and the plan for implementing those architectures.
- In highly distributed computing, a layered architecture is recommended for a simple reason to allocate the different responsibilities of the system to the respective layers.
- With the same principle, Enterprise Architecture is built on the same layer design concept. It is inspired with the idea to execute the relevant processes and services of the layer and its related components.

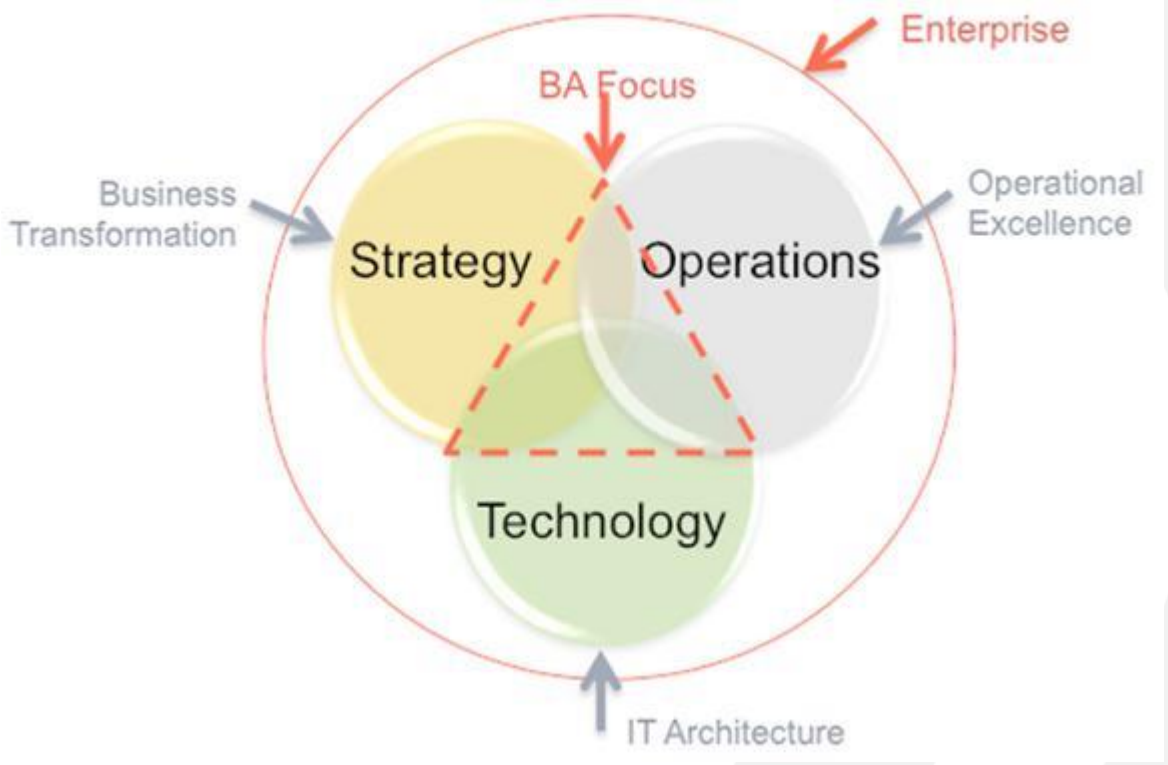
Architecture segregation



Business Architecture

- Business Architecture is nothing but a blueprint of the enterprise.
- It helps you understand the organization and supports you to align the long-term strategic objectives and short-term tactical demands.
- Basically, it is the bridge between enterprise strategy and business functionality

Business Architecture



Key aspect by Business Architecture

S.No.	Query raised	Target delivery	Sample
1	Who?	Stakeholders	End customers, senior managers.
2	Why?	Strategy Tactics	Vision, mission, objectives of the business.
3	How?	Initiatives Projects	Innovative development Operational excellence.
4	When?	Events	Business critical moments.
5	What?	Products Services	Manufacturing output Customer facingservice.
6	Where?	Policies Regulations	Governing body Corporate policy Company process.
7	How well?	Metric Measurement	Financial report Revenue trend Profit sharing.

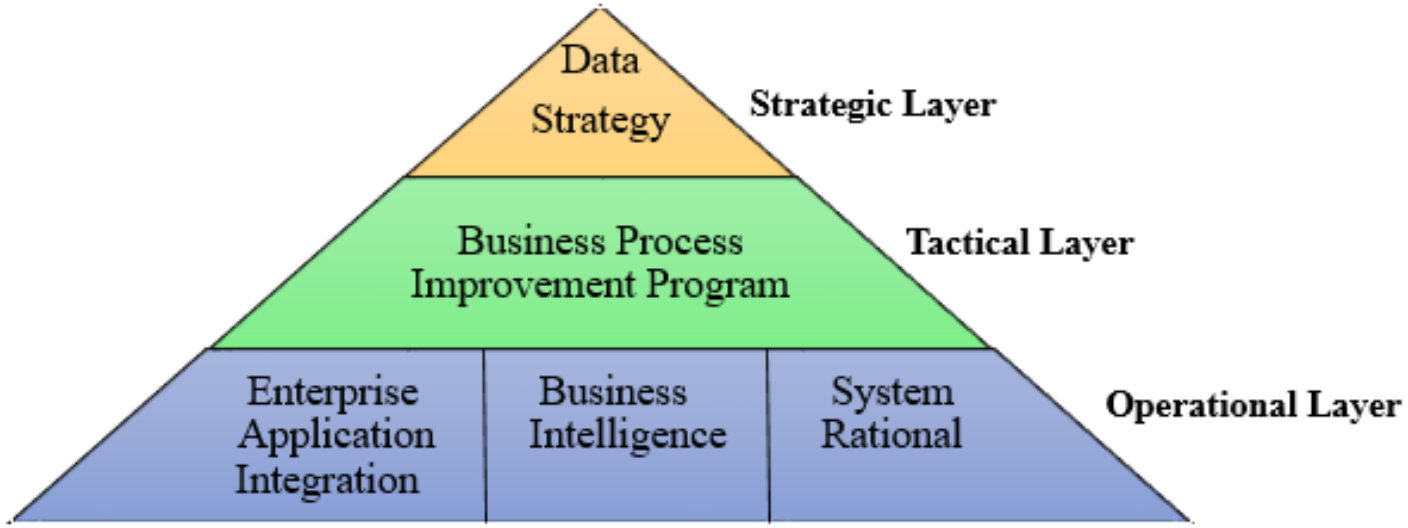
Business Architecture case studies

- An Oracle Business Architecture Case Study, December 2012, Transforming Revenue Management at Brink's
- Available at https://www.oracle.com/technetwork/articles/entarch/oea-case-study-brinks-2012-1883032.pdf

Data Architecture

- Data Architecture is the key contract between the business and technology in an enterprise.
- Data Architecture is designed in such a way that the real business data is handled smoothly across the layers of the enterprise architecture.
- It plays the key role/artifact to develop and implement governance supporting the enterprise data strategy.
- It collaborates/connects with the various enterprise objects, such as hardware, applications, networks, tools, technology choices, and data.

The framework/layers of the Data Architecture



The framework/layers of the Data Architecture

- Most of the ground-level operations are executed in the lower components--Enterprise Application Integration (EAI), Business Intelligence (BI), and System Rationalization.
- Data is tactically architecture at the middle layer using the BPI program.
- The top layer of Data Architecture is getting involved in the Data Strategy of the underlying enterprise.

Application Architecture

- Software application is designed to meet an organizational need.
- As the business model is quite common in a similar industry, it obviously expects the software application to build with the common architecture to satisfy the business requirements of an enterprise.
- As a result, Application Architecture is built in a generic way to create the business system, which is required to meet the specific requirements of the business.

Application Architecture

- Application Architecture specifies the leveraging technologies.
- Technologies are easily used to implement information systems, such as data, processes, and interfaces.
- On top of that, Application Architecture describes the details of the internal components and the way they interact to build the complete information system.
- Application Architecture exhibits the execution steps and methods in the model of the system blueprint into the reality of the leveraging enterprise.

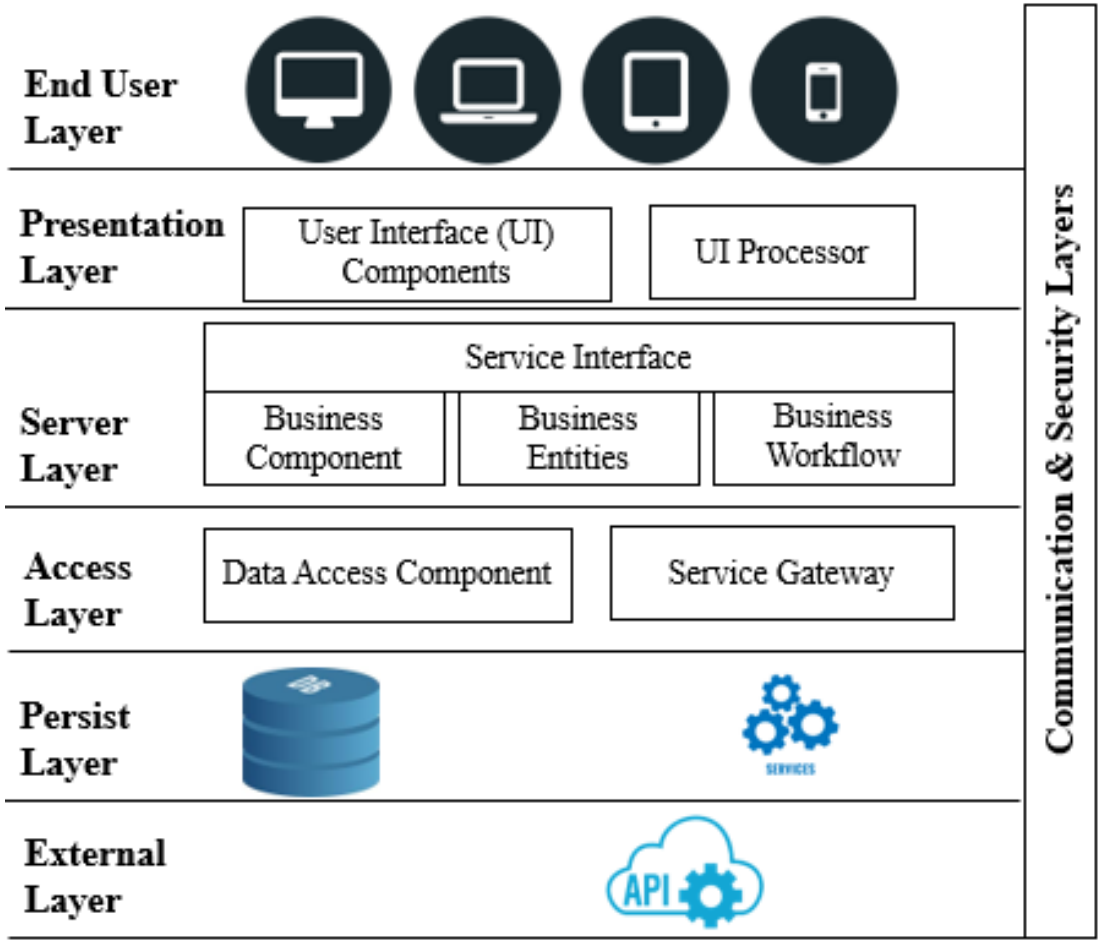
Application Architecture

- Application Architecture is designed into the logical groupings of the software components.
- These logical layers help you differentiate between the different kinds of tasks performed by the components. In turn, the system is easier to support the design principle of reusability across the platform.

Application Architecture

- Layers vs. Tier.
  - Layer describes the logical groupings of the functionality/components in an application.
  - Tier describes the physical distribution of the functionality/components on the underlying hardware systems.
- Each layer can be implemented as a large-scale component running on a separate server. It is the most used web-based architectural model in the industry.

Application Architecture



Six layers

- End User Layer
  - This is an individual who uses the product after it is fully developed and marketed.
  - The End User Layer is essential to build for desktop, web, mobile, pad, and so on.
- Presentation Layer:
  - This contains the end user oriented functionality responsible for managing user interaction with the core system.
  - It consists of components that provide a common bridge between the end user and core business logic

Six layers

- Server Layer:
  - The core functionality of the target system by encapsulating the relevant business logic.
  - Consists of components, some of which may expose service interfaces that other callers can use. It is termed as the heart of the system.
- Access Layer
  - Is a bridge between the core business/server layer and the persisted store layer.
  - It is designed using the best access pattern and practices of the enterprise architecture. It has two key components, namely the Data Access Component (DAC) and Service Gateway (SG).

Six layers

- Persistence Layer:
  - It provides access to data hosted within the boundaries of the system and data exposed by other networked systems.
  - By design, data is accessed through services in modern Application Architecture.
- External Layer:
  - Is designed to expose the functionality of the application as the services to the external customer.
  - API is the popular term in the industry, through which business services are exposed externally to earn the profit by sharing the best services.

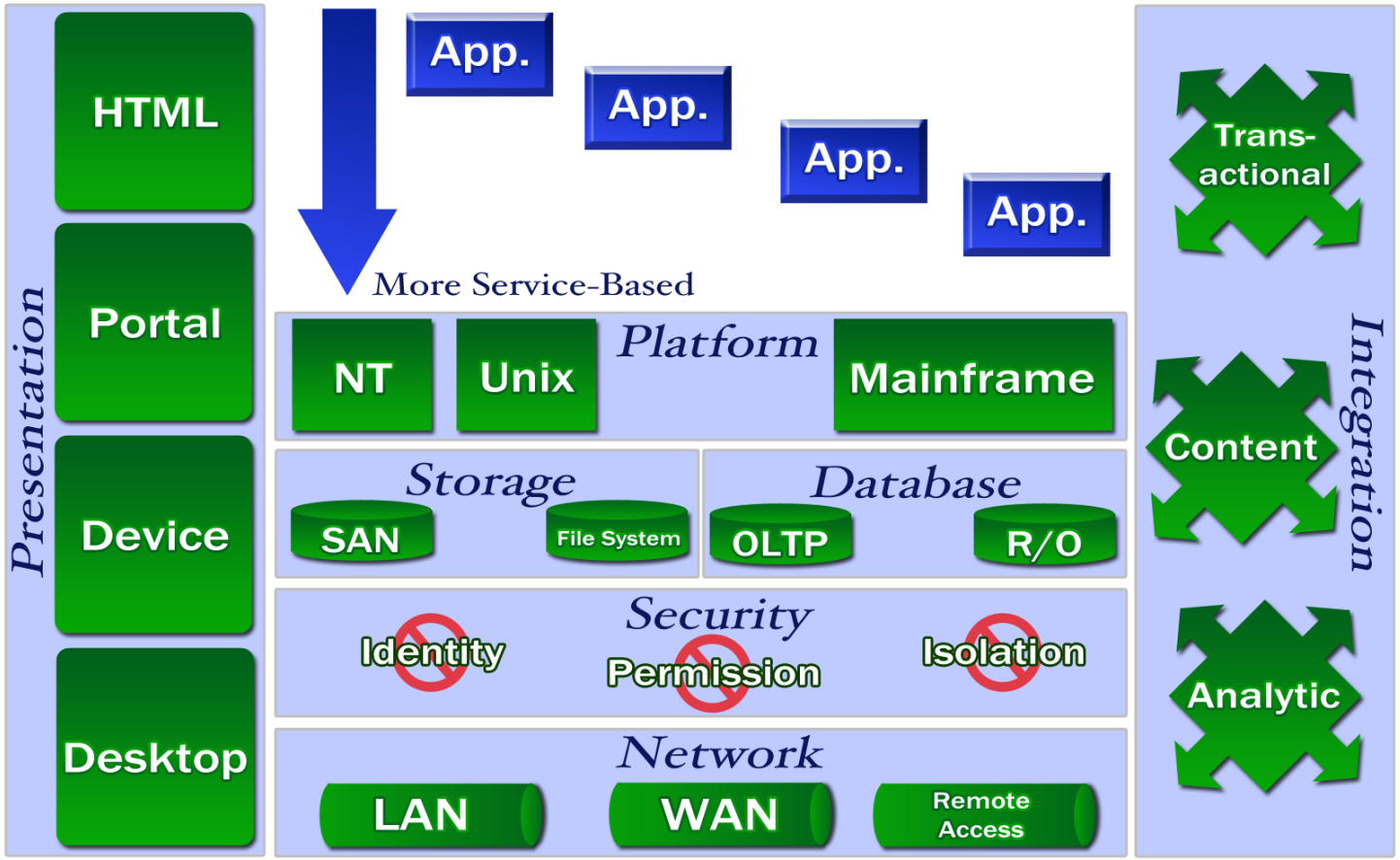
Applications Architecture

- Is the art and science of ensuring the suite of enterprise applications to create the composite architecture
- The characteristics
  - Scalability
  - Reliability
  - Availability
  - Manageability.

Technology Architecture

- Are defined in collaboration with operational staff.
- It is the duty of the application architect to correct any wrong assumptions that the team might make regarding enterprise infra-architecture.
- It covers the servers, desktops, storage, network, and so on.
- It is easy to adapt into the latest trends instead of reinventing the wheel again on our own.

Technology Architecture



The Open System Interconnection (OSI) model

		OSI Model	Examples
Data	7	**Application Layer** Facilitates communiation between software applications like Outlook, IE	Web Application
Data	6	**Presentation Layer** Data representation and encryption	HTTP
Data	5	**Session Layer** Interhost communication	80
Segments	4	**Transport Layer** End to end connection and reliability	Transmission Control Protocol (TCP)
Packets	3	**Network Layer** Path determination and logical addressing	Internet Protocol (IP)
Frames	2	**Data Link Layer** Mac and LLC - Physical addressing	Ethernet
Bits	1	**Physical Layer** Media, signal and binary transmission	CAT5

The Open System Interconnection (OSI) model

- Is an interesting area in Technology Architecture.
- It defines a networking framework to implement protocols in seven layers.
- As the lower/deep technical details, control is passed from one layer to the next, starting from the Application Layer to the bottom most Physical Layer of bits.
- The communication passes over the channel to the next station and back up the hierarchy.

Summary

- Enterprise Architecture (EA) is expected to have the responsibility to perform strategic steps in alignment with business vision.
- EA has few strong fundamental blocks, namely agility, durability, efficiency, and effectiveness.
- EA is the discipline of addressing business needs with people, process, and technology, with the definition of the purpose, intent, and structure of any system.

Reference

- Senthilvel, G., Khan, O. M. A., & Qureshi, H. A. (2017). Enterprise Application Architecture with. NET Core. Packt Publishing Ltd.

Q&A

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